Building a Spectroscopy Apparatus Around the FP4209 InstaTune

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by

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Contents

Acknowledgments

Mom, Dad, your roommate, maybe even your thesis advisor – give credit where credit's due! For my part, I'd like to thank the Math department for making such a nice LaTeX template that we could use as a base for our Physics template.

Introduction

Methane and Climate Change

Problems in Modeling Methane's Behavior

Ro-Vibrational Transitions and HITRAN Data

In addressing climate change, we must frequently take measurements of greenhouse gasses in Earth's atmosphere. To measure the amount of methane in the atmosphere, we must first have a model of the intensity of the absorption spectrum of methane at different temperatures and pressures. Uncertainties in these models, particularly in lineshape functions, propagate to uncertainties in greenhouse gas measurements.

Spectroscopic models of greenhouse gasses used in measurement of the atmosphere typically neglect accounting for collisional effects that lead to broadening and narrowing effects of the line shape of the spectra produced by these gasses.

We take our measurements with a novel digitally controlled IR laser spectrometer created using the FP4209 InstaTune laser module with the intent to characterize the effects of pressure on the line shape function used in modeling the predicted absorption spectrum of methane. The FP4209 scans over a IR wavelength range of 1627.5000 – 1672.4955 nm, which corresponds very well with the ro-vibrational absorption spectrum of methane. Interpolating DAC values and scanning through the range of wavelengths produced by this laser allows us to construct a quasi-continous profile of the spectrum of the intensity of light absorbed by the methane gas.

This work allows us to take new measurements of the properties of methane gas in a controlled environment, to better understand the spectra we observe in the atmosphere, and to more precisely describe the behavior and amount of methane in our atmosphere.

CHAPTER 1

Theory

Laser Spectroscopy

You can write text in LaTeX as you would in any other word processing or typesetting software. You can incorporate *different* **varieties** of *typeface*. What's really exciting about writing text in LaTeX is that it's very easy to incorporate weird symbols, like α or $\otimes$ or ∞ into your text. Note that the weird symbols appear in between $ signs – this means that they occur in *math mode*.

You create a new paragraph just by putting in a blank line. Leaving extra space or extra lines in between paragraphs won't change their spacing at all. LaTeX will automatically indent your paragraphs, so you don't have to worry about indenting.
You can, however prevent it from indenting your paragraphs if you'd like. But that's getting a little ahead of ourselves.

Laser Mechanics

It is beautifully easy to write equations, whether they occur inline with the text $\Psi(x,t) = Ae^{i(kx-\omega t)}$ or as a separate line

$$\Psi(x,t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}. \tag{1.1}$$

To continue a paragraph after an equation, make sure that there's no blank line between the equation and your next line of text. Equations that are in a separate line will be numbered, and it's easy to refer to Eq. 1.2 by labelling your equations

$$i\hbar \frac{\partial}{\partial t} \Psi(x,t) = \hat{H} \Psi(x,t). \tag{1.2}$$

Are you getting tired of writing begin{eqnarray} and end{eqnarray} yet?[1] You can simplify your life by defining abbreviated symbols for commands:

$$\langle x | \Psi(t) \rangle = \Psi(x,t). \tag{1.3}$$

For simplicity, I'd suggest putting all of your new commands in a separate include file.

When writing a derivation with multiple steps, it is often useful to use an equation array

$$\begin{aligned} \int_0^{2\pi} \sin^2 x dx &= \int_0^{2\pi} \frac{1}{2} \left(1 - \cos 2x\right) dx \\ &= \frac{1}{2} \left(x - \frac{\sin 2x}{2} \right) \Bigg|_0^{2\pi} \\ &= \pi. \end{aligned}$$

[1] Notice how I made a } by putting a backslash in front of the symbol. The curly brace has special meaning in LaTeX, so if you want to make one, you have to let the program know that you want the symbol.

Note that you can suppress equation numbering, which is often desirable in derivations!

The syntax for equation arrays is similar to the syntax for arrays within equations. For example,

$$\hat{H} = \begin{pmatrix} \Delta & \Omega & 0 \\ \Omega^* & 0 & \kappa \\ 0 & \kappa^* & -\Delta \end{pmatrix} \tag{1.4}$$

There are many more complicated things you can do with equations, but this should be enough to get started!

Distributed Bragg Reflector (DBR) Lasers

If you want to make a table, you'll need to specify the justification for each of the columns; TEXwill automatically calculate the column widths for you.

Name	Age	Town	Time
Joe Schmoe	23	Rumford	1:04
Mary Q. Contrary	44	Yarmouth	1:12
Baxter Boulevard	32	Portland	1:42
Lass Plaice	60	Lewiston	1:57

TABLE 1. Not too many people showed up for the first, and probably last, annual Androscoggin swimming race.

You might also, for example, want to

(1) make lists
(2) number your lists

- use bullet points
- find better items to put in your bullet points

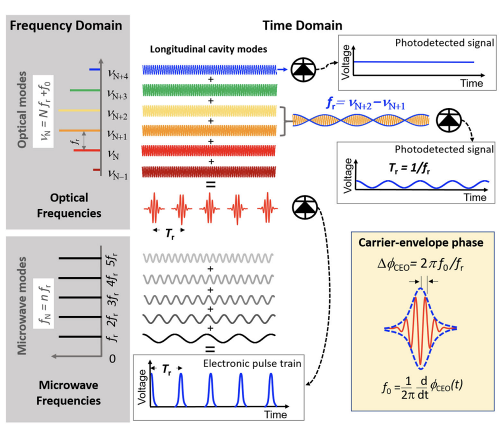


FIGURE 1.1. This is a very bad figure.

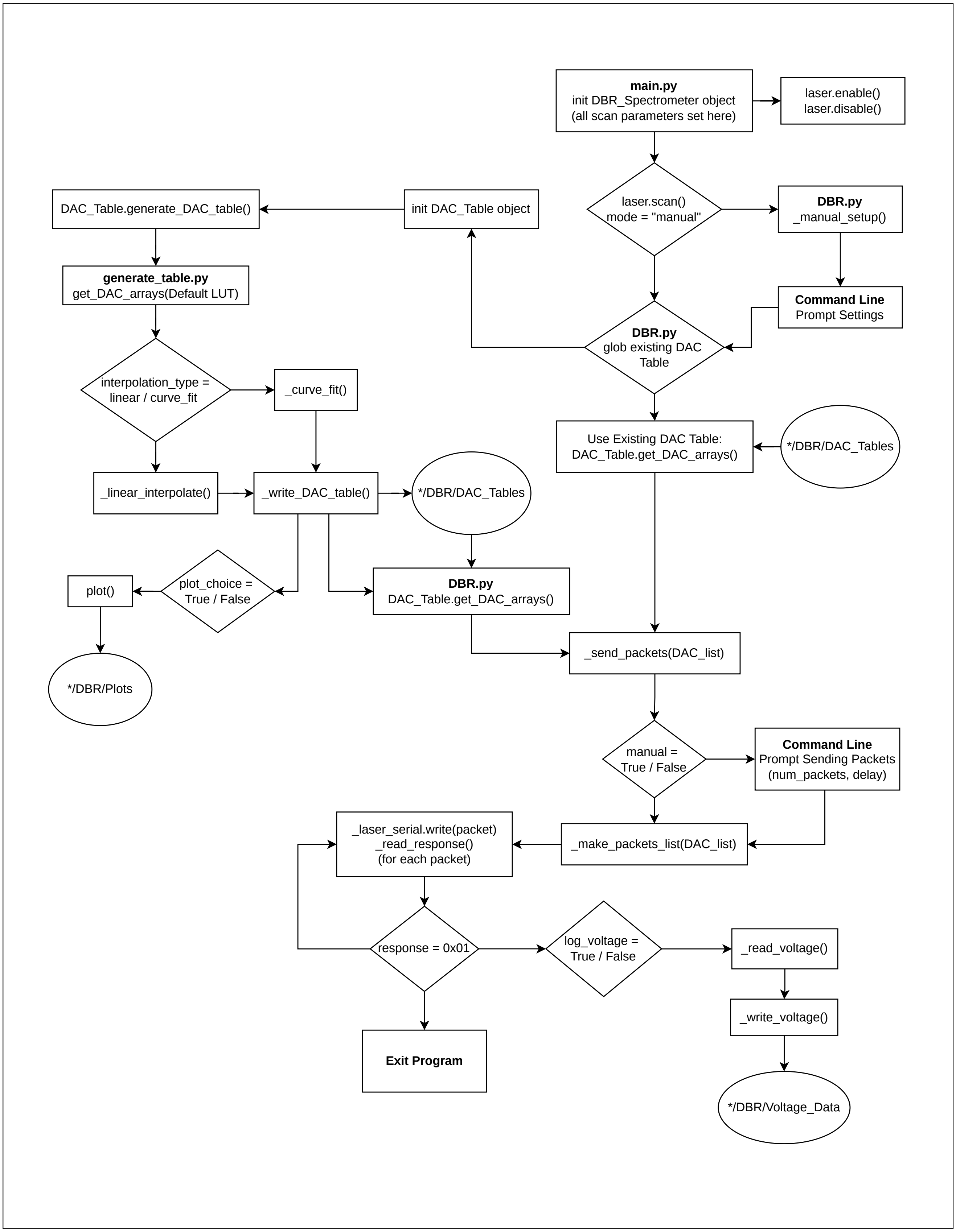
test citation [1]

CHAPTER 2

Method

Laser Functionality

Program Sequencing



Communication Protocol

Data Collection

WOOOOO

CHAPTER 3

Analysis

DAC LUT Discontinuities

Laser Precision

Spectroscopic Measurements

wooooo

Appendix A

Long derivations, marginally relevant details, and the like can be relegated to your appendix, which unlike the mammalian counterpart will never require removal.

Bibliography

[1] D.E. Chang et al. "Quantum optics with surface plasmons". In: *Physical Review Letters* 97 (2006), p. 053002.